

ALTUS QUARTZ MT, OK, USA

WMO: 720354

Lat: 34.699N

Lon: 99.338W

Elev: 437

StdP: 96.19

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 63901

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.3	-6.1	-16.2	1.0	-4.2	-13.4	1.2	-1.9	14.7	6.9	12.8	7.7	4.2	10	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	40.1	21.3	38.2	21.7	37.0	21.7	24.9	33.1	24.2	32.5	23.7	31.8	5.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.4	27.9	22.2	17.8	27.4	21.5	17.1	27.0	77.9	33.0	75.3	32.5	73.1	31.8	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4)	12.6	11.2	9.9	DB	-12.5	41.0	2.7	2.1	-14.4	42.6	-16.0	43.8	-17.5	45.0	-19.4	46.6
(5)				WB	-13.7	25.5	2.6	1.0	-15.5	26.3	-17.1	26.9	-18.5	27.4	-20.4	28.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	5.2	7.3	12.9	16.9	22.4	28.0	29.2	28.6	24.4	17.7	11.2	5.5
	DBStd	9.81	5.11	6.12	5.64	5.06	4.55	2.99	2.77	2.85	3.50	5.07	5.07	4.91
	HDD10.0	521	163	111	36	8	0	0	0	0	0	6	44	152
	HDD18.3	1691	407	311	183	84	16	0	0	0	3	74	217	397
	CDD10.0	3256	14	36	124	215	384	540	594	578	431	245	82	13
	CDD18.3	1384	0	2	13	41	141	290	336	319	183	55	4	0
	CDH23.3	17310	12	46	186	539	1607	3732	4567	4068	1868	602	78	5
	CDH26.7	9327	1	11	48	211	779	2138	2706	2336	868	212	17	0
Wind	WSAvg	4.5	4.3	4.8	5.3	5.5	5.2	5.2	4.0	3.7	3.9	4.3	4.2	4.1
Precipitation	PrecAvg	686	25	28	45	68	94	94	65	67	59	61	42	31
	PrecMax	1027	71	105	114	202	277	217	198	188	180	150	166	104
	PrecMin	426	0	0	0	2	4	1	5	0	4	1	0	0
	PrecStd	166	20	27	32	45	60	59	43	50	40	37	43	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	28.0	31.0	35.1	38.4	41.3	42.2	42.5	37.5	33.5	29.1	23.0
		MCWB	12.8	13.6	15.4	17.7	18.5	20.6	20.9	20.9	21.2	19.1	15.7	12.7
	2%	DB	20.9	24.0	27.6	31.8	36.1	38.9	40.0	39.8	35.0	31.3	25.7	19.6
		MCWB	10.1	12.8	15.7	17.5	20.2	21.3	21.8	21.4	21.6	19.3	15.1	11.5
	5%	DB	17.5	20.9	25.2	28.9	33.8	37.3	38.3	37.6	33.1	29.0	22.7	16.3
		MCWB	8.8	11.6	15.2	17.1	19.6	21.6	21.9	22.1	21.7	18.7	14.0	9.4
	10%	DB	14.7	17.6	22.5	26.3	31.3	35.7	36.7	35.9	31.7	27.0	20.0	13.8
		MCWB	7.5	10.3	13.9	16.5	19.5	21.8	22.1	22.4	21.4	17.4	13.1	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	16.7	19.5	21.2	23.7	25.2	25.7	25.8	24.8	23.0	18.5	16.2
		MCDB	21.2	21.5	25.8	28.1	30.6	33.8	33.2	33.6	31.8	29.0	23.2	19.3
	2%	WB	11.4	14.7	18.5	19.9	22.9	24.1	24.7	25.0	23.9	21.8	16.8	12.8
		MCDB	18.1	21.1	24.3	26.7	30.4	33.0	32.8	33.5	30.8	27.7	22.0	16.7
	5%	WB	9.5	12.8	17.1	19.1	22.1	23.6	24.1	24.2	23.1	20.5	15.5	10.5
		MCDB	16.5	19.5	21.8	25.5	29.4	32.3	32.5	32.6	29.5	25.8	20.5	15.6
	10%	WB	7.8	10.7	15.5	18.0	21.2	22.9	23.5	23.7	22.5	19.1	14.1	8.4
		MCDB	13.9	17.2	20.5	23.8	28.4	31.6	31.8	31.6	28.7	25.1	19.4	13.7
Mean Daily Temperature Range	5% DB	MDBR	14.0	14.1	14.2	14.1	13.2	13.0	12.8	12.6	12.2	13.8	14.1	12.0
		MCDBR	19.7	19.7	17.5	18.6	16.6	15.3	15.2	14.8	14.2	16.0	17.9	17.1
		MCWBR	10.9	10.3	8.5	7.7	5.4	3.8	3.4	4.1	4.6	6.5	9.0	9.9
	5% WB	MCDBR	18.1	17.5	13.6	14.4	13.2	13.0	12.4	12.2	11.5	12.8	14.2	14.8
		MCWBR	10.5	10.3	7.6	7.4	5.6	4.4	3.9	4.3	4.4	6.5	8.5	9.5
Clear-Sky Solar Irradiance	taub	0.282	0.297	0.331	0.364	0.385	0.412	0.415	0.405	0.373	0.342	0.311	0.284	
	taud	2.537	2.496	2.420	2.351	2.367	2.326	2.357	2.333	2.398	2.473	2.489	2.530	
	Ebn at Noon	937	955	943	920	900	872	866	871	886	891	889	912	
	Edh at Noon	80	93	109	123	123	128	123	124	111	94	83	76	
All-Sky Solar Radiation	RadAvg	3.04	3.66	4.87	5.93	6.41	7.05	7.02	6.38	5.33	4.09	3.17	2.59	
	RadStd	0.30	0.48	0.31	0.46	0.66	0.58	0.39	0.37	0.35	0.54	0.39	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page